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數據科學專題

Term Project Final Report

第 19 組

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1、簡介 （Introduction）

● 研究背景（Background）

Lymph node metastasis is one of the most important indicators for determining cancer stage and planning subsequent treatment. In current clinical practice, pathologists still mainly rely on manually examining high-resolution H&E-stained slides, carefully scanning grid by grid to identify suspicious tumor regions. This workflow is time-consuming and, under prolonged reading, prone to fatigue and inter-observer variability. With the increasing adoption of digital pathology, how to leverage deep learning to assist diagnosis and reduce missed lesions has become a very popular research topic in recent years.

In our project, we use the Kaggle Histopathologic Cancer Detection dataset, which consists of 96×96-pixel lymph node patches cropped from the Camelyon series of whole-slide images, with a total of around 220,000 images. The label of each image indicates whether the central region of the patch contains metastatic cancer cells, so the task can be formulated as a binary classification problem: the input is an H&E patch, and the output is the predicted probability of “metastatic” vs. “non-metastatic.”

● 研究動機（Motivation）

We chose this topic because, on the one hand, we wanted to implement an image classification task that is truly related to a real medical scenario, and on the other hand we hoped to gain a deeper understanding of how stain variation affects model performance. While reviewing the literature, we found that many papers mention that stain normalization (such as the Macenko method) can reduce color differences between hospitals; meanwhile, many other works emphasize the benefits of data augmentation for improving robustness. However, on a single public dataset like HCD, the conclusions about which design is actually more helpful are not very consistent. This made us curious: if we systematically compare “stain normalization” and “color augmentation” under the same backbone and training pipeline, can we obtain clearer insights?

- 研究目標（Research Aims）

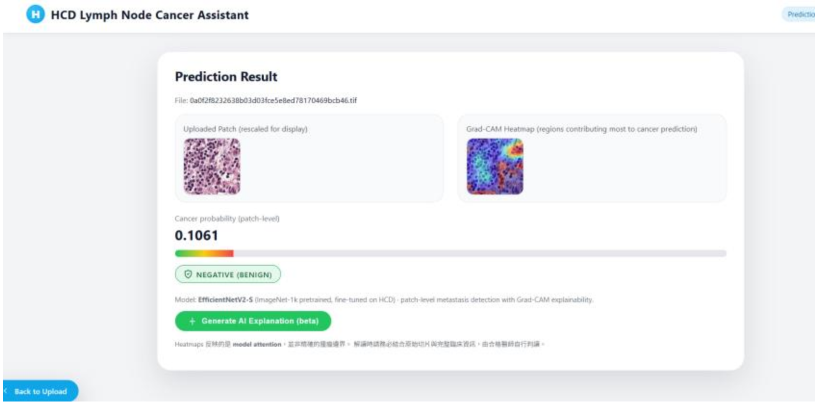


Figure 1

Based on the above background and motivation, the main goals of this project can be summarized into three points:

- Re-implement and compare existing models: Under a fair stratified 70/15/15 split, we reproduce the ResNet-18 baseline and ResNet-34 SOTA settings from the HCD paper, and use them as the baselines for our later experiments.

- Evaluate the effect of stain normalization and color augmentation: Using EfficientNetV2-S as the main backbone, we design a 2×2 experiment with {Raw, Macenko} × {Color aug ON, Color aug OFF}, and analyze how AUROC, F1-score, and other metrics change under different combinations, as well as the interaction between the two factors.

- Deploy the best model as an assistive tool: We package the best-performing EfficientNetV2-S model into a small web-based system (as illustrated in Figure 1) that allows users to upload patches, shows prediction probabilities and Grad-CAM heatmaps, and generates Chinese explanations via an LLM, in order to simulate possible future clinical usage.

- 挑戰與貢獻（Challenges and Contributions）

During the project, the main challenges we faced were the large data volume, high training cost, and the many confounding factors in stain design that are difficult to disentangle at once. The HCD dataset contains about 220,000 patches. To compare multiple models (ResNet-18/34 and EfficientNetV2-S) under several color settings, we had to automate and parallelize the training pipeline in order to finish all experiments within one semester. Therefore, we implemented an 8-GPU DDP training pipeline with mixed precision to speed up training. In addition, although HCD does exhibit stain variation, it ultimately comes from a single challenge, so there is still a gap compared with real cross-hospital domain shift. How to present our conclusions clearly in the report and avoid over-claiming was also something we needed to handle carefully.

Under these limitations, our contributions have three main points.

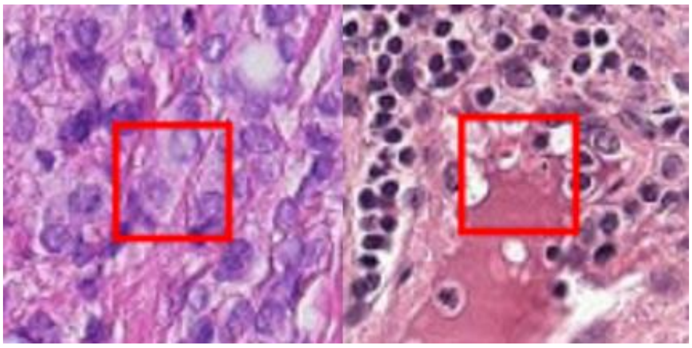
- First, on the same stratified split, we re-organized and compared ResNet18, ResNet34, and EfficientNetV2-S. We found that the newer backbone does clearly better than the older ones in both AUROC and F1. In particular, recall increases from about 0.94 to 0.97, which is very important for not missing metastatic lesions.

- Second, with a 2×2 experiment design, we found that "Raw HCD + moderate color augmentation" is actually the most stable setting. Macenko stain normalization slightly reduces the performance on this dataset, which is a bit different from our first intuition but is supported by the experiments.

- Third, we deploy the best patch classifier as a web demo that includes Grad-CAM visualization and AI-generated explanations in Chinese.

2、問題設定 （Problem Setting）

● 問題描述 （Problem Description）

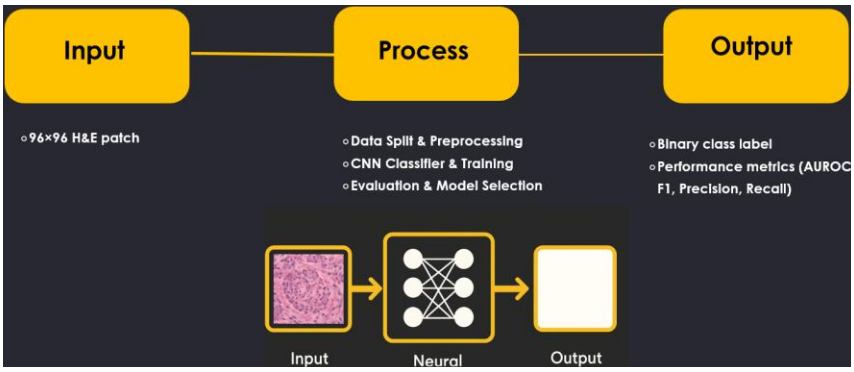


metastatic vs. non-metastatic

In this project, the main problem we want to solve is: given a small lymph node H&E patch (96×96), can a model automatically decide whether there are metastatic cancer cells inside or not? We use the Kaggle Histopathologic Cancer Detection (HCD) dataset, which contains about 220,000 96×96 images. Each image is cropped from a lymph node whole-slide image and labeled as a binary task: the central region either has metastasis (1) or does not have metastasis (0).

If we describe it using “Input / Process / Output”:

- Input: a preprocessed lymph node H&E patch. The raw image may first go through (optional) Macenko stain normalization, then be resized to 256×256 and normalized.
- Process: we feed these patches into our deep learning models for training, including the ResNet18/34 baselines from the original paper and our main model EfficientNetV2-S. During training, we compare four settings: Raw vs. Macenko × with / without color augmentation.
- Output: for each patch, the model gives a predicted probability of metastasis and a Positive / Negative classification result. When we deploy it as a web demo, we also generate Grad-CAM heatmaps to help pathologists see which regions the model is focusing on.



Input / Process / Output

● 評估指標

In this task, we deal with a slightly imbalanced binary classification problem: the proportion of positive (metastatic) patches is relatively small. In the clinical setting, missing a cancer case is more serious than flagging a few extra suspicious patches, so we do not only look at overall accuracy. Instead, we use AUROC, F1-score, precision, and recall as our main evaluation metrics.

AUROC (Area Under the ROC Curve) can be understood as the model’s overall ability to rank positive samples ahead of negative samples across all possible thresholds, which makes it a fair metric when we compare different models or different color–stain settings.

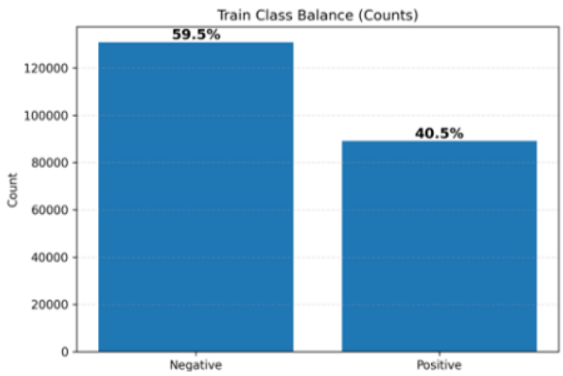
Precision means: among all patches the model predicts as cancer, what proportion are actually cancer.
Recall means: among all true cancer patches, how many are successfully detected by the model.
The F1-score is the harmonic mean of precision and recall, and gives a balanced trade-off between these two measures.

Evaluation metrics

3、方法設計 （Proposed Methods）

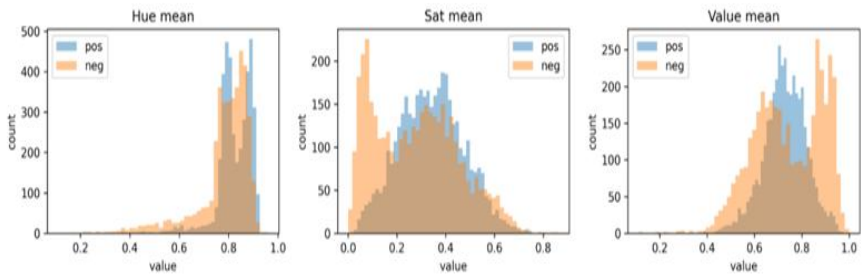
- Data Understanding -- Challenge 1

Before building our models, we first did some basic EDA on the Kaggle Histopathologic Cancer Detection (HCD) dataset. More specifically, we plotted the class distribution of the train / validation / test splits, and we can see that the data is slightly imbalanced: about 60% are negative and 40% are positive.

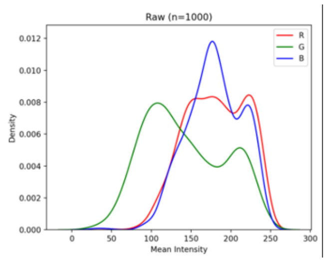


Class balance bar chart

Next, for each 96×96 patch, we computed the average HSV values and compared the positive and negative patches on the hue, saturation, and value channels. We found that the hue distributions almost completely overlap, but the brightness and saturation distributions are slightly shifted between the two classes. Finally, we also plotted kernel density curves for the three RGB channels and observed clear stain intensity variation overall, as well as shifts in the R/G/B channels.



Hsv mean per patch distribution



RGB channel KDE

These observations make us more confident that, although the stain variation in HCD comes from the same challenge, there are still clear differences in brightness and intensity. If we do not handle these differences, the model might rely too much on certain specific color patterns. Because of this, in our later experiments we include both Macenko stain normalization and color augmentation, and systematically evaluate which approach is more suitable for this dataset.

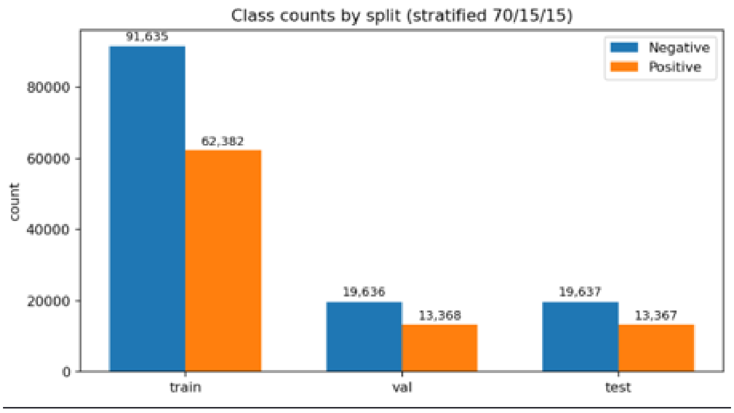
- Data Preparation-- Challenge 1

For all experiments, the original 96×96 HCD patches are first resized to 256×256 using bicubic interpolation before being fed into EfficientNetV2-S. There are two reasons for this choice:

(1) the ImageNet-pretrained EfficientNetV2-S was originally trained around 224/256 resolution, so we would like to keep a similar receptive field;

(2) a higher input resolution makes the Grad-CAM heatmaps visually smoother and easier for pathologists to inspect.

For data preprocessing, we first apply a stratified 70 / 15 / 15 split to the original 220k patches. This makes sure that the positive/negative ratio in the train, validation, and test sets is similar, and helps reduce instability in the evaluation metrics caused by sampling.



stratified 70 / 15 / 15 split

Next, we designed a 2×2 experimental setup to study how to handle stain variation:

- one dimension is Raw HCD vs. Macenko-normalized;
- the other dimension is with vs. without color augmentation.

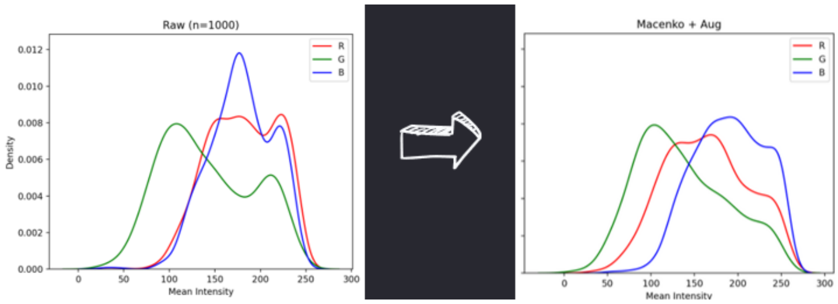
For stain normalization, we use the Macenko method. We first convert the RGB image into optical density (OD) space, then estimate the stain basis for hematoxylin and eosin. After that, we project each patch onto this basis and re-normalize the stain intensities, and finally reconstruct an RGB image with more consistent colors. In theory, this step can reduce the variation between slides caused by different staining intensities, and it is one of our approaches to handle Challenge 1.

In addition, we designed two color augmentation strategies:

- For the raw HCD images, with 90% probability we randomly choose one of three augmentation branches, and apply medium-strength changes in brightness, contrast, and gamma, and sometimes add a small RGB shift. The remaining ~10% of patches keep their original colors, serving as anchors so the model still sees the "real" staining.
- For the Macenko-normalized images, since the colors have already been standardized, we only apply relatively mild

brightness/contrast/gamma changes, in order not to break the color tones that Macenko has already aligned.

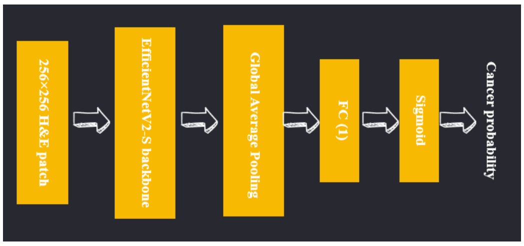
This preprocessing design allows us to fairly compare "Raw vs. Macenko" and "with vs. without color augmentation" under the same training recipe. It directly corresponds to Challenge 1: systematically analyzing how stain and color design affect model performance on a single public dataset.



Stain norm. & Color enhancement

- Data Modeling-- Challenge 2

For the model design, we wanted to check whether a newer backbone can really outperform ResNet34 under a reasonable training cost. So, we chose EfficientNetV2-S as our main model, using ImageNet-1k pretrained weights. The model input is a 256×256 H&E patch. After passing through the EfficientNetV2-S backbone to extract features, we add a Global Average Pooling layer and a fully connected layer with output dimension 1, then apply a **sigmoid** function to get the metastasis probability. During training, we use the AdamW optimizer with a cosine learning rate schedule, and a total batch size of about 1.8k (8 GPUs, 64 per GPU with gradient accumulation). We train for 12 epochs. We also adopt a class-balanced BCEWithLogitsLoss, giving slightly higher weight to positive samples to reduce the impact of class imbalance. In addition, we enable mixed precision and channels-last to speed up training.For comparison with previous work, we re-implement the original paper's ResNet18 baseline and ResNet34 SOTA, and train them on the same stratified 70/15/15 split. This setup allows us to directly compare the advantages of EfficientNetV2-S in terms of AUROC, F1, and recall, and systematically answer Challenge 2: under the same data and experimental settings, does a modern backbone really bring a stable performance improvement?

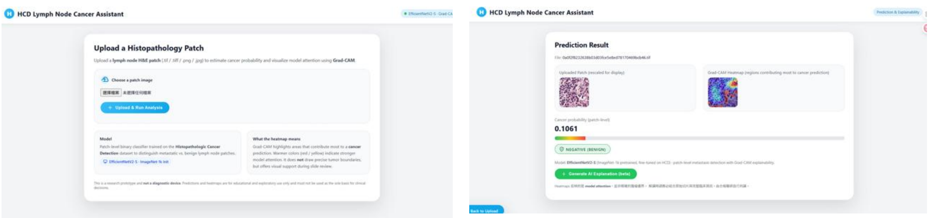


Model structure

- Deployment-- Challenge 3

In addition to comparing metrics in our offline experiments, we also wanted the model to be a bit closer to a tool that could actually be used in the clinic, so we built a simple web-based assistant. On the backend, the system uses the EfficientNetV2-S model we trained earlier. When a user uploads a lymph node H&E patch on the website, the server automatically resizes and normalizes the image, and then runs inference. The frontend shows:

- the patch-level cancer probability and the Positive / Negative prediction,
- the corresponding Grad-CAM heatmap, which highlights the regions the model mainly focuses on,
- a Chinese explanation generated by the Google Gemini API.



web-based assistant

4、實驗設定 （Experimental Setting）

- 資料集相關描述（如：資料欄位、型態、總筆數、年份等）

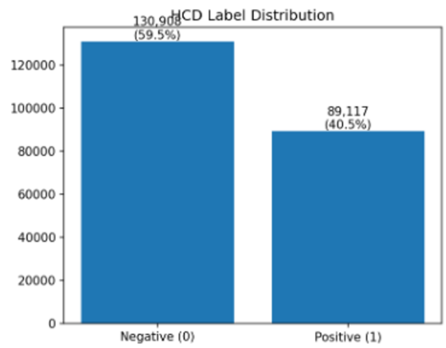
Source Link: https://www.kaggle.com/competitions/histopathologic-cancer-detection

Dataset Checklist

- V Category Limitation：Healthcare & Biomedicine

- V Data Source：Histopathologic Cancer Detection (HCD)
- V Sample Size：> 220,000 patches
- V Feature Quality & Interpretability：H&E image pixels provide direct medical semantics
- V Suitable for Classification Tasks：Binary classification (tumor present: 1/0)

In this project, we use the Histopathologic Cancer Detection dataset from Kaggle.
HCD is built from lymph node H&E images collected for the CAMELYON challenge. Each sample is a 96×96 pixel patch with a binary label: **1** means the patch contains metastatic breast cancer cells (metastasis), and **0** means there are no tumor cells. In total, there are about 220,000 patches, and the ratio of positive to negative samples is roughly **4:6**.



label distribution

Baseline Models

In this project, both our baseline and SOTA models directly follow the settings used on the Kaggle Histopathologic Cancer Detection dataset in the paper by Wang et al., "ResNet for Histopathologic Cancer Detection, the Deeper, the Better?".

- Baseline: ResNet-18

We first treat the ResNet-18 model in the paper as a *strong baseline*. The reasons are: ResNet-18 has fewer parameters, a simpler architecture, and is a very common backbone in medical imaging. Also, previous work has already shown that it can achieve good AUROC and F1 performance on the HCD dataset.

In our implementation, we follow the paper's setting: we use ImageNet pretrained weights, replace the last fully connected layer with a binary classification head, and then fully fine-tune the model on the 96×96 HCD patches.

- State-of-the-Art: ResNet-34

In the same paper, under the HCD dataset with full fine-tuning, ResNet-34 reaches an AUROC of about 0.992, which is the best among ResNet-18/34/50/152. Therefore, we treat it as our SOTA model.

In our experiments, we use the same preprocessing, data split, and training pipeline as for ResNet-18, and only replace the backbone with ResNet-34. This lets us see whether our proposed EfficientNetV2-S pipeline can match or even surpass this SOTA model under a similar training cost.

- 評估策略（Evaluation Strategy）

We use a single stratified 70 / 15 / 15 hold-out split to evaluate model performance. More specifically, we first do a stratified random split based on the labels (metastatic / benign): about 70% of the patches are used as the training set, 15% as the validation set, and 15% as the test set. We also fix the random seed so that all models use exactly the same data split, making the comparison fair.

During training, we use the validation AUROC as the main metric for model selection. After each epoch, we compute AUROC on the validation set and save the checkpoint with the best performance. At the end, we only take this best model and evaluate it on the test set. In addition, the decision threshold is also tuned on the validation set (for example, choosing the threshold that gives the best F1-score), and we keep this threshold fixed when testing.

In the test phase, we report AUROC, F1-score, precision, and recall. AUROC reflects the overall separability between positive and negative samples; F1-score balances precision and recall. In the clinical setting, we especially care about not missing cancer cases, so we also look at recall separately. All baseline models (ResNet-18 and ResNet-34 from the paper) and our EfficientNetV2-S models follow exactly the same data split and evaluation pipeline, so the comparison between different methods is consistent.

- 實驗環境（Environment）

Hardware

- 8-GPU(DPP)
- Each GPU with enough VRAM for batch size 64 (per GPU)
- Mixed-precision training (AMP) to speed up training

Software

- OS: Ubuntu Linux (server)
- Programming language: Python 3.10
- Frameworks: PyTorch
- Data augmentation: albumentations
- Metrics & utilities: scikit-learn, pandas, numpy
- Development: VS Code

5、實驗結果（Experimental Results）

- 實驗結果（需有圖表，以及針對圖表的闡釋）

- Overall Performance and Comparison with Baseline Models

Model / Setting	AUROC	F1-score	Precision	Recall
ResNet18 baseline	0.9889	0.9419	0.9455	0.9383
ResNet34 (paper SOTA)	0.9897	0.9449	0.9505	0.9393
EfficientNetV2-S (ours, raw + color aug)	**0.9957**	**0.9694**	**0.9710**	**0.9678**

Experiment Results Table

First, we compare the overall performance of different backbones under the same data split. The result table summarizes the performance of ResNet-18 (baseline) and ResNet-34 (the best setting in the reference paper), as well as our EfficientNetV2-S model on the Kaggle

Histopathologic Cancer Detection dataset. For a fair comparison, we re-implemented ResNet-18/34 following the original paper: using 90×90 random crop, horizontal/vertical flips, and ImageNet normalization. The only change is that we replaced the original random 8:1:1 split with a stratified 70/15/15 split, so that the positive/negative ratio is consistent across train/val/test.

From the table, we can see that under the same training pipeline, EfficientNetV2-S clearly outperforms both ResNet-18 and ResNet-34 in terms of AUROC and F1-score. In our experiments, ResNet-34 reaches around 0.97 in both AUROC and F1, while EfficientNetV2-S can further improve these scores. In particular, recall increases from about 0.94 to around 0.97, meaning the model is less likely to miss metastatic patches under the same decision threshold. In a decision-support setting, we would rather flag more suspicious patches for the pathologist to review than miss true metastases, so this improvement in recall is especially important to us.

- Ablation Study on Color Augmentation and Stain Normalization

	Color aug OFF	Color aug ON
Raw HCD	AUROC 0.9955, F1 0.9676	**AUROC 0.9957, F1 0.9694**
Macenko	AUROC 0.9854, F1 0.9346	AUROC 0.9855, F1 0.9342

Stain Normalization × Color Augmentation

Next, we focus on how different color processing strategies affect the model. To separate the effects of each factor, we design a 2×2 experiment: {Raw HCD, Macenko-normalized} × {Color augmentation ON, OFF}, and compare the AUROC and F1 of these four combinations while keeping all other training settings fixed.

- Raw + Color Augmentation

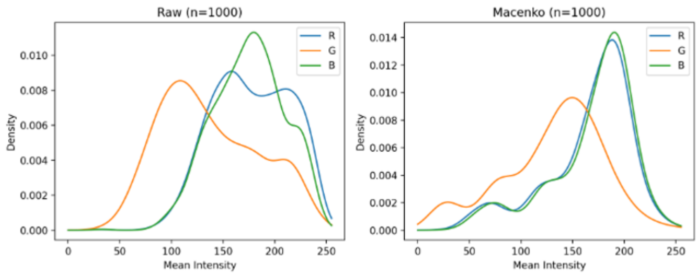
This setting is the best among the four. The F1 score increases from about 0.9676 without color augmentation to about 0.9694 with it. Although the difference in numbers is small, it is still a stable advantage at the scale of 220,000 patches. This suggests that, on top of the original HCD color distribution, adding moderate brightness/contrast/gamma perturbations can make the model less sensitive to realistic stain variations.

- Color augmentation has limited impact but provides a stable regularization effect

Our raw HCD dataset already has quite large stain variation and a lot of data, so even without color augmentation, EfficientNetV2-S can already learn fairly stable features. Color augmentation mainly plays the role of regularization – it helps prevent the model from memorizing specific stain styles too much, rather than giving a huge performance boost. This also explains why the scores only improve slightly, instead of showing a very dramatic difference.

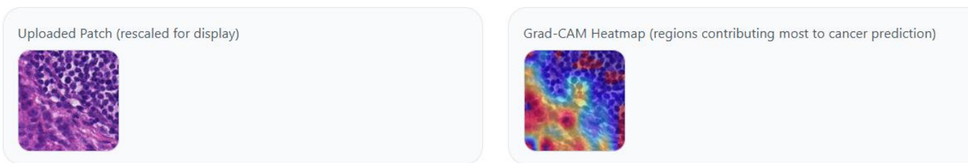
- All Macenko setting performance worse than the Raw setting

In contrast, no matter whether we use color augmentation or not, once we apply Macenko stain normalization first, both AUROC and F1 become clearly lower than with raw HCD, mostly around 0.93–0.94. Combined with our EDA on the HSV/RGB distributions, we guess that for a dataset like HCD, which comes from a single challenge with relatively consistent scanning protocols, the original stain variation actually carries useful information. When Macenko "flattens" the colors of different slides into a very similar style, it also removes some local contrast that helps distinguish metastasis vs. non-metastasis, so the overall performance drops slightly.



Raw RGB vs Macenko RGB

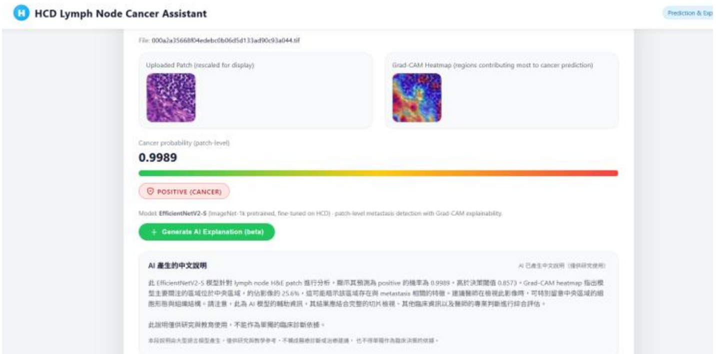
- Grad-CAM Visualization and Web Demo



Raw patch and grad-cam heatmap

In addition to the numerical metrics, we also use Grad-CAM to visualize where the model is focusing on each patch. The figure shows a few representative examples: the left side is the original H&E patch, and the right side is the Grad-CAM heatmap overlay. We can see that for metastatic samples, the model tends to focus on regions with a higher density of tumor cells, and the hot areas mostly fall in places that a pathologist would also pay attention to. For benign samples, the highlighted regions are more scattered or weaker.

Based on this model, we implemented a simple web-based assistant for single-patch uploads. After the user uploads a lymph node patch, the system outputs: (1) the patch-level cancer probability and the Positive/Negative prediction, (2) the Grad-CAM heatmap, and (3) a Chinese explanation generated by a large language model. This part does not directly contribute to the quantitative scores, but it helps us check whether the model behaves reasonably on real images and also makes it easier to discuss possible real-world usage with clinicians in the future.



Demo

- Insights

• Modern backbone does bring real improvements

Under exactly the same data split and training pipeline, EfficientNetV2-S consistently performs better than ResNet-18 and ResNet-34, especially in terms of recall, which directly helps with the goal of “not missing cancer cases.” This shows that in medical imaging tasks, newer-generation architectures still have clear value in both parameter efficiency and feature representation.

• Moderate color augmentation is useful, but not the main factor

Adding medium-strength color augmentation on the raw HCD dataset gives a stable but limited improvement in F1. So we treat it as a “necessary but not the most important” design choice: it helps the model become less sensitive to small stain changes within the same domain, but the factors that really make a big difference are still the improved backbone and the amount of data.

• On single-source data, aggressive stain normalization can backfire

Macenko was originally designed to handle stain variation across different slides and hospitals. But for a dataset like HCD, where the source is relatively uniform, forcing all images to match one reference style can accidentally remove subtle contrast cues between tumor and non-tumor regions. This reminds us that stain normalization does not always “make things better” by default — we still need to consider the actual data source and task requirements.

6、總結與限制 （Conclusion and Limitation）

- 總結過程遇到的難題，以及對應之解決方法。

During this project, we actually ran into quite a few practical difficulties.

The first problem was training stability.
The HCD dataset has more than 220,000 96×96 patches, and the class distribution is slightly imbalanced. When we first trained EfficientNetV2-S with DDP on 8 GPUs, the loss often went up and down a lot, and sometimes different GPUs were not in sync.
Later, we made training more stable and reproducible by tuning the batch size, learning rate, and AMP precision settings, and by fixing all the details such as data loading and random seeds. After that, the results of each run became much more consistent, and the metrics were easier to reproduce.

The second major challenge was how to design color augmentation and stain normalization.
At the beginning, we tried applying very strong color augmentation to all patches, but the training became very noisy, and the validation AUROC / F1 were both unsatisfactory. We then switched to our current design: about 90% of patches go through one of three branches with brightness / contrast / gamma changes and a small RGB shift, while the remaining 10% are almost unchanged and serve as anchors with the original colors. This compromise lets the model see enough color variation without making the images look too unrealistic.

For Macenko stain normalization, we also spent quite a lot of time on estimating the stain basis offline and doing batch processing. In the end, we built a preprocessing pipeline that is stable and can be reproduced reliably.

- 總結研究成果（包含實驗分析、洞見分析、研究貢獻等）。

Overall, our main findings can be divided into three parts.

First, for the backbone comparison, we successfully applied EfficientNetV2-S to the HCD task and compared it with our reproduced ResNet-18 baseline and the ResNet-34 SOTA model from the paper. Under the same stratified 70/15/15 split, EfficientNetV2-S achieves slightly higher AUROC and F1 than both ResNet models. In particular, recall can be improved from about 0.94 to 0.97, which suggests that the newer architecture is better at capturing fine tissue textures and is less likely to miss metastasis cases.

Second, in the color-related ablation study, we found that medium-strength color augmentation on raw HCD does help, but the gain is limited. Although HCD comes from two hospitals, both belong to the same CAMELYON challenge, and the staining protocols are relatively similar. In our HSV / RGB EDA, we also saw that stain variation exists, but it is not as severe as a typical cross-hospital domain shift. As a result, EfficientNetV2-S can already learn fairly stable representations even without augmentation, and moderate color augmentation mainly plays a regularization role, increasing F1 only slightly from 0.9676 to 0.9694.

Third, we observed that Macenko stain normalization actually hurts performance on this dataset. In our experiments, all four Macenko-based combinations reach AUROC / F1 scores of about 0.93–0.94, which is clearly lower than the raw H&E settings. Combined with our color-distribution EDA, we hypothesize that in a single-challenge dataset, the original stain intensity and contrast already contain useful cues for separating tumor and non-tumor regions. Macenko heavily “flattens” the color differences between slides and may wash out some of these helpful details at the same time. This result reminds us that stain normalization is not automatically “the stronger, the better” — it has to be chosen based on the task and data source.

Besides the quantitative metrics, we also wrapped our best EfficientNetV2-S model into a simple web demo. Users can upload a single patch, and the system outputs the patch-level cancer probability and the Positive / Negative prediction, along with a Grad-CAM heatmap and a short Chinese explanation. This demo does not directly affect our scores, but for teaching and presentation purposes, it helps us more easily check whether the model’s attention on real images is reasonable, and it is also a small step toward more interpretable medical AI.

- 描述此研究的限制（Limitation），以及未來發展（Future Works）。

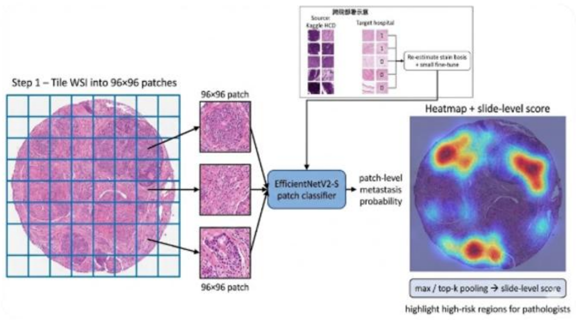
This project still has several limitations. First, we only use a single public dataset, HCD, and adopt one fixed stratified 70/15/15 split. We do not perform full k-fold cross-validation and do not include any external cohort. Therefore, our conclusions strictly speaking only reflect what happens "under the HCD setting", and are not enough to guarantee that the same effects will hold in other hospitals or under different staining protocols.

Second, we mainly compare a few supervised backbones (ResNet-18/34 and EfficientNetV2-S), and have not yet introduced architectures that are closer to real WSI tasks, such as self-supervised pretraining or multiple instance learning (MIL). In addition, for the color augmentation design, hyperparameters like "keeping 10% of patches in original colors" have not been thoroughly tuned with systematic grid search or cross-validation to find the optimal combination.

For future work, we have two relatively clear directions.

The first direction is cross-hospital adaptation (domain adaptation). Our idea is to treat the current EfficientNetV2-S trained on HCD as a pretrained model. When moving to a new hospital, we can use a small number of labeled patches from that hospital to re-estimate the stain basis or do a light fine-tuning, then re-evaluate AUROC / F1 on the external cohort and adjust the decision threshold. The goal is to ensure that the sensitivity remains high enough in the new hospital setting.

The second direction is to answer a question raised by classmates in class: "How can the ability of small patches be extended to whole-slide images (WSI)?" Our planned approach is: at a fixed magnification, we cut the WSI into many 96×96 patches, apply our current EfficientNetV2-S to each patch to obtain patch-level metastasis probabilities, then map these probabilities back to their original coordinates to form a slide-level probability map / heatmap. Finally, we use something like max pooling or top-k pooling to aggregate the patch scores into a single slide-level positive probability. In this way, we can both provide pathologists with a map highlighting high-risk areas and truly extend the model from patch-level to the WSI workflow. In the future, if we can obtain real WSI data and multi-center data, we hope to implement this pipeline as an extension of this project and as a more practical next step toward deployment.



Patch-level to WSI

7、參考文獻 (Reference)

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8、組員分工與各自執行細項 (Work Distribution Chart)

	林呈軒	許晴鈞
Survey	V	V
Data Understanding	V	V
Data Preparation	V	V
Data Modeling	V	V
Evaluation	V	V